EXP NO:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

AIM:
To upload and analyze the data set given in csv format and perform data preprocessing and visualization.

PROGRAM:

python

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Read the csv dataset
df = pd.read_csv(
    'https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv'
)
df

# First five rows
df.head()

# Information about the dataset
df.info()

# Statistical description
df.describe()

# Check missing values
df.isnull().sum()

# ----- Data preprocessing -----

# Remove duplicate rows
df = df.drop_duplicates()

# Drop the column with too many missing values
df = df.drop(columns=['deck'])

# Fill missing numerical values with median
df['age'] = df['age'].fillna(df['age'].median())

# Fill missing categorical values with mode
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])
df['embark_town'] = df['embark_town'].fillna(df['embark_town'].mode()[0])

# Confirm no missing values remain
df.isnull().sum()

# Encode categorical column
df['sex_code'] = df['sex'].map({'male': 0, 'female': 1})
df[['sex', 'sex_code']].head()

# ----- Data visualization -----

# Countplot of survivors
sns.countplot(x='survived', data=df)
plt.title("Survived (0 = No, 1 = Yes)")
plt.show()

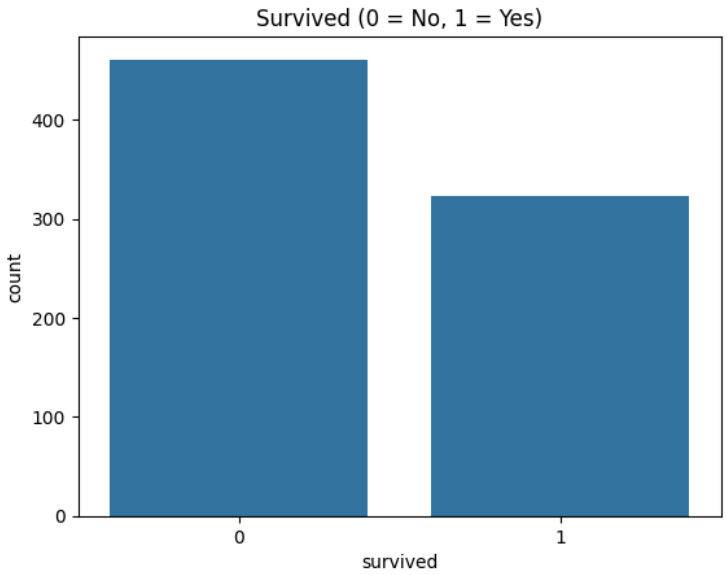
# Survival by gender
sns.countplot(x='sex', hue='survived', data=df)
plt.title("Survival by Gender")
plt.show()

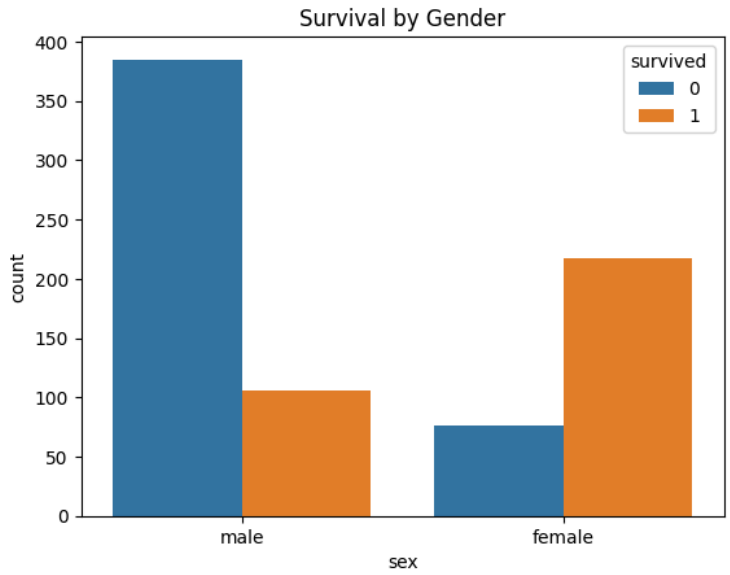
# Histogram of age
sns.histplot(df['age'], kde=True)
plt.title("Age Distribution")
plt.show()

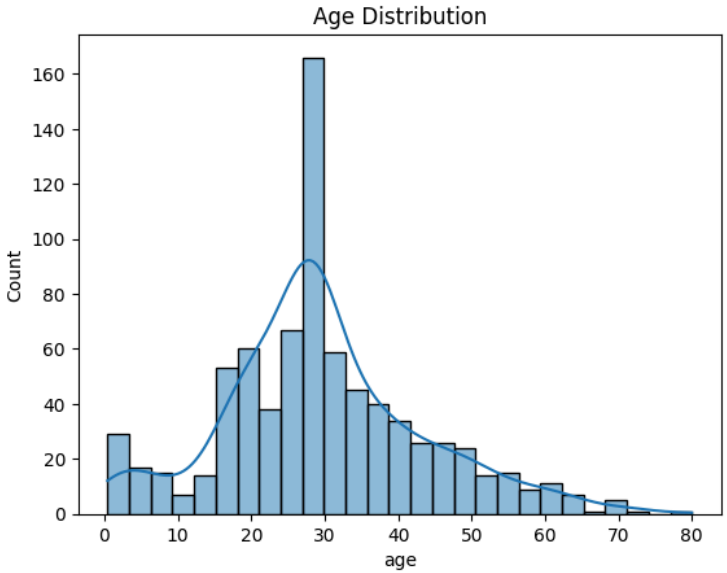
# Boxplot of fare by class
sns.boxplot(x='pclass', y='fare', data=df)
plt.title("Fare by Passenger Class")
plt.show()

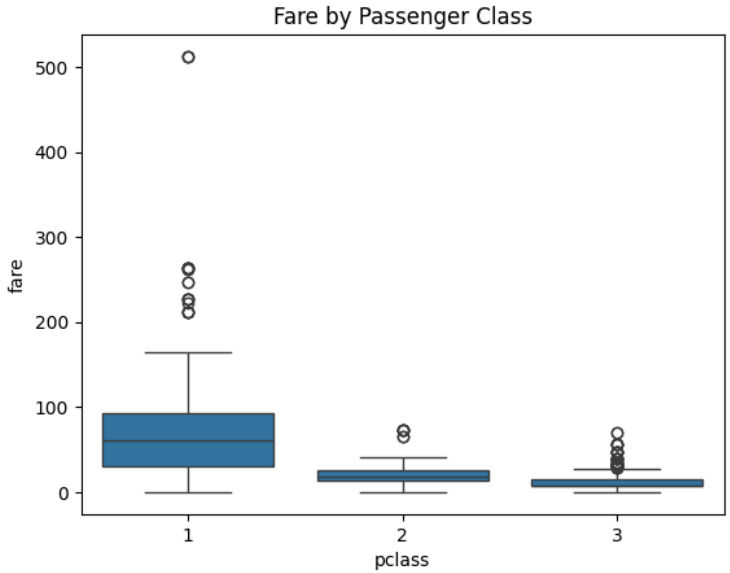
# Correlation heatmap
sns.heatmap(df[['survived', 'pclass', 'age', 'fare']].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

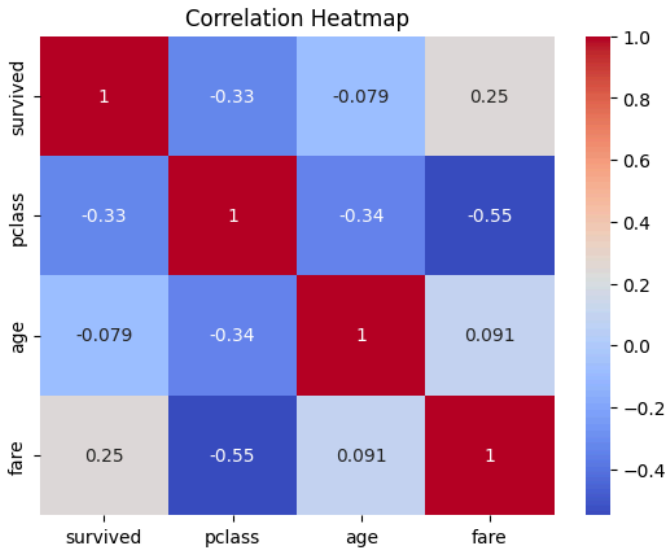
OUTPUT:


Survived (0 = No, 1 = Yes)
count
survived
0
1
0
100
200
300
400


Survival by Gender
survived
0
1
count
sex
male
female
0
50
100
150
200
250
300
350
400


Age Distribution
Count
age
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100
120
140
160


Fare by Passenger Class
fare
pclass
0
100
200
300
400
500
1
2
3




```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    object
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    object
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), float64(2), int64(4), object(7)
memory usage: 92.4+ KB
```

RESULT:

Thus the program to upload and analyze the data set in csv format and perform data preprocessing and visualization was executed successfully.